STRAT-TRACE

Source-tagged ^{7}Be for process-aware aerosol model evaluation

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1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

Central question. Which changes in upper-atmospheric supply and wet removal can surface ^{7}Be concentration and deposition records distinguish, and when does improved model agreement conceal compensating errors?

Air-quality models must reproduce not only concentrations, but also the processes determining aerosol persistence and deposition. STRAT-TRACE will use the existing beryllium-7 (^{7}Be) implementation in the Danish Eulerian Hemispheric Model (DEHM) to build a controlled test of process credibility. One modelling system, one radionuclide and a small, fixed experiment set will connect source-resolved budgets to the information contained in real measurements. The intended advance is a **regime-resolved discrimination benchmark**: evidence showing which process explanations are supported, excluded or observationally indistinguishable.

For particle-bound ^{7}Be, production, boundary supply, transport, radioactive decay and deposition jointly determine the measured signal [1–3]. Surface air and deposition provide complementary constraints, while production uncertainty and precipitation regime affect their interpretation [1–4]. A source label is an accounting tool, not an additional observation. STRAT-TRACE therefore tests whether model-derived differences survive the actual sampling, uncertainty and independent evaluation, rather than equating sensitivity with identification.

State of the art and the specific advance

Established capability	What STRAT-TRACE will add
Joint ^{7}Be concentration–deposition evaluation [1,2] and improved production formulations [3].	A controlled test of when joint observations reject compensating supply–removal explanations, including nuisance-source uncertainty.
Source tagging already exists; GEOS-Chem includes a stratospheric-source Be7s tracer [17].	A verified multi-origin budget linked to discriminatory power at the actual measurement intervals and its dependence on observing regime.
Controlled removal contrasts [5] and transport-model comparisons [6].	A compact within-DEHM intervention set with identical meteorology and blocked evaluation, yielding a protocol reusable without a second model.

Table 1. Novelty is the integrated, observation-aware test of process explanations; individual tracer and statistical tools are established.

The scientific payoff extends beyond one model's calibration: the benchmark will specify which measurement combinations and regimes are informative for aerosol-process evaluation. The operational relevance is concrete: DEHM contributes to the European CAMS regional air-quality system [18], providing a route for developers to assess the diagnostic method before considering operational changes.

Measurable objectives, hypotheses and success criteria

Objective / delivery	Testable hypothesis and evidence
O1 — Source-resolved response By M9: verified source contributions and budgets.	**H1:** wet-removal changes produce origin-dependent responses. Test source-fraction and absolute-loss contrasts across seasons and years. Reject differential sensitivity where responses are indistinguishable relative to numerical and sampling uncertainty.
O2 — Discrimination of explanations By M15: supply–removal response library.	**H2:** joint air–deposition observations discriminate some scenarios that air-only evaluation cannot. Compare recovery at the same controlled false-discrimination rate; report where adding deposition yields no resolvable gain.
O3 — Transfer and observing value By M21: held-out tests and discrimination map.	**H3:** budget-supported explanations retain joint skill across held-out years and regimes. Reject transfer when gains disappear, uncertainty becomes excessive, or one observable improves by worsening the other.

These hypotheses do not presuppose that scavenging is the only deficient process or that descent is too rapid. Upper production, imposed boundary supply, lateral input and measurement representativeness remain alternatives. Success requires a verified response library and a reproducible classification of its observational support; it does not require every regime to identify a unique mechanism. Failure to discriminate will be accompanied by a quantified detection bound and a test of what additional observing information would help.

Preparation: existing implementation and measurement access

Chham and Ye's existing collaboration has implemented ^{7}Be in DEHM. Their EGU 2026 abstract documents production mapping, WRF/ERA5 meteorology, aerosol-bound treatment and surface evaluation [7]. The applicant's earlier resolution analysis provides a concrete process question [16]:

DEHM: 25 km relative to 75 km	Median response
Surface ^{7}Be concentration	+5.9%
Bulk deposition: wet + dry	+13.6%
Precipitation	−45.9%

Table 2. Preliminary JJA comparison, 2009–2024, 79 locations [16]. Each response is 100 × (fine/coarse − 1) from paired seasonal means, followed by a spatial median. Locations need not be independent cells. These preparatory configuration contrasts motivate the new experiments; they do not establish observation biases or faster descent.

Greater deposition with lower precipitation motivates testing tracer availability and removal exposure; it does not establish faster transport or a faulty scavenging coefficient. The new work moves from this ambiguity to controlled interventions and observational tests, remaining distinct from the existing implementation/evaluation manuscript.

The applicant confirms 2009–2024 as the currently available experimental deposition baseline; acquisition of additional years is under way [16]. DWD correspondence documents co-located air/deposition sampling and daily precipitation since 2009 [21]. Availability is distinguished from station-level completeness: sample dates, collector changes, gaps and permissions will be inventoried. New records will undergo the same quality control before incorporation. Core delivery does not depend on receipt of additional years.

Scope protected for a 24-month fellowship

The core uses DEHM alone, existing meteorology and existing ^{7}Be measurements. Its reduced-chemistry configuration and already available computational access support long-period experiments without adding model systems. No new campaign, second radionuclide, GEOS-Chem, HYSPLIT or altered vertical winds are required. Longer runs test interannual stability, not climate-change attribution.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

One physical tracer; transparent source and boundary accounting

The reference is the team's existing reduced-chemistry DEHM–WRF/ERA5 workflow at 75 km, with 29 layers and a top near 100 hPa [7,16]. The experiment retains transport, mixing, physical decay and wet/dry deposition; reduced chemistry is not a reason to omit these processes. The code revision, process operators, aerosol treatment, CRAC-based production [9], geophysical inputs and boundaries will be documented and frozen. Layer 1 is the surface; layer numbers increase upward. Existing computational access enables an extended common simulation period.

Production P(x,t) is partitioned with non-negative masks w_k such that $\Sigma_k w_k = 1$ and $P_k = w_k P$. All labels retain the same transport, physical decay and deposition. Labels record where material enters the simulation, not its subsequent altitude or last tropopause crossing.

Label	Source partition and interpretation
UPPER / MIDDLE / LOWER	Production at p < 300 hPa; 300 ≤ p < 700 hPa; and p ≥ 700 hPa, respectively, within atmospheric cells. These are diagnostic bands, not three separate meteorological systems.
TOP / LATERAL	Actual external upper and lateral inputs, separately accounted for. Boundary concentrations are converted to the realised transfer; they are not assumed to prescribe constant mass input.
INITIAL	Initial inventory propagated through spin-up. Its residual influence is tracked rather than silently attributed to subsequent production.

Table 3. Six labels plus an independently computed total tracer. Pressure cut-offs are pre-specified design choices; a short boundary-shift test will check sensitivity to them.

UPPER and TOP cannot be called the complete stratospheric contribution: DEHM does not represent the full stratosphere, and pressure alone is not a source-origin observation. Imported material retains a boundary label unless its upstream history is available. Nested-boundary accounting must not double count previously tagged material.

Numerical verification and process budgets

Source partition is not mass-balance closure. The sum of label concentrations, deposition and burdens must reconstruct the independent total. Numerical limiters and boundary updates will be tested explicitly, including reconstruction after a source reweighting. The proposed acceptance target is a relative reconstruction error below 0.1% above a declared absolute floor, and at least ten times smaller than the smallest interpreted response; this is a test criterion, not an achieved result.

For each label k and common control volume, an atom inventory N_k will be checked against:

$$dN_k/dt = P_k + F_{in,k} - F_{out,k} - \lambda N_k - Lwet_{,k} - Ldry_{,k} + \varepsilon_k.$$

P and F are production and transfer rates, L denotes removal, λ is the physical decay constant and ε is the residual. Fluxes and operator losses will be accumulated at model time steps; pressure-band budgets include moving-boundary/rebinning transfers. Whole-domain closure is mandatory. Uninstrumented local terms will be reported as unresolved rather than inferred from concentration gradients.

Primary outputs are absolute C and D, source fractions C_k/Ctotal and D_k/Dtotal with denominator floors, and integrated wet losses by origin and pressure band. The removal-rate diagnostic ∫Lwet dt / ∫N dt has inverse-time units, not the meaning of a deposited fraction. Source-reweighted contributions are used only where linear reconstruction has passed; missing boundary and storage terms cannot be absorbed into "transport".

A fixed experiment matrix across an extended common period

2009–2024 (16 years) is the available experimental deposition baseline, not a fixed study limit [16]. The reference and both wet-removal settings will share the same meteorology and full period. Longer simulations require consistent archived inputs; additional deposition years now being obtained will extend joint evaluation after quality control. The baseline supports core delivery without relying on unreceived records. Model-input coverage and usable station intervals will be inventoried separately; no complete station network or dates beyond the baseline are assumed.

Case	Intervention	Coverage / interpretation
R + tags	Unchanged ^{7}Be production and reference process operators.	Common window anchored to 2009–2024; extendable with consistent inputs. Source attribution and reference budgets.
W− / W+ + tags	Multiply the reference aerosol wet-removal rate coefficients by $\alpha w = 0.5 / 2$; reference $\alpha w = 1$.	Both settings span the full common period. Fixed stress contrasts, not confidence limits or calibrated improvements.
S contrasts	Reweight UPPER and TOP contributions separately within each removal setting, after linearity tests.	Full common period by matched-tag reconstruction. Supply changes; meteorology and transport operators do not.
Verification controls	Short source-linearity, band-boundary and spin-up convergence checks.	Bounded pilots, not an expanding catalogue of scientific cases.

Table 4. Baseline budget: 3 configurations × 16 years = 48 configuration-years, not a maximum. A common window of N years requires 3N configuration-years, plus spin-up and verification; tags add species cost.

The wet intervention scales rate coefficients before the existing time integration, not the deposition output afterwards. Actual removed mass must be transferred to the deposition ledger with positivity maintained. The source masks, physical decay, dry deposition, meteorology and transport operators stay fixed. This deliberately tests the strength of represented wet removal; it does not identify a unique deficient scavenging sub-process. The implementation point and test log are pre-submission readiness items.

Supply contrasts retain uncertainty in global production and lateral input as nuisance alternatives. A scalar change in supply is not a change in transport speed. For each wet setting the correct label fields are used: reweighting reference fields cannot emulate a different wet-removal run. Independent source rescaling is diagnostic; plausible uncertainty scenarios must preserve documented dependencies between production and boundary forcing. A supply–removal interaction is a difference of differences in a common output, expressing changing exposure to the sink.

Interannual robustness and independent temporal evaluation

Before tracer fitting, seasons will be classified by rainfall amount, wet-day frequency, event persistence and available mixing diagnostics. The extended record tests response stability across wet/dry years and contrasting meteorological regimes. Modelled and observation-supported coverage will be reported separately; unsupported regimes receive no empirical generalisation claim. Daily rainfall supports daily/weekly metrics, not sub-hourly intensity. Longer records broaden the test but cannot resolve structurally confounded responses or establish climate attribution.

Whole-year, multi-year blocks will be assigned to selection and locked evaluation using meteorology, coverage and collector-method eras before tracer fitting. All three configurations cover both sets; evaluation blocks remain excluded from fitting. Records received after protocol lock enter a pre-registered temporal extension without retuning. Spatial groups account for shared model cells; whole-year resampling tests year and regime dependence. High-frequency 3-D output is restricted to selected episodes; receptor series and budgets span the full simulation.

Measurements, discrimination tests and interpretation

DWD records will be linked through station history, sample dates, relocations and collection method [21]. The reported 2018 collector transition will be retained as a method stratum: pre-/post-transition results and stable-method subsets will be checked separately, not pooled as a homogeneous record. Only quality-controlled paired samples enter joint tests; air-only stations provide context. Panel size and usable years come from the inventory, not the 79 preliminary model locations.

The observation operator averages air activity over collection windows, flow-weighted when possible, in mBq m^{-3}, and integrates deposition over its own windows in Bq m^{-2}. Air and deposition windows need not be identical: each is simulated on its actual support. DWD confirms that collector rinsing includes dry material, so these data require a **bulk-deposition** comparison [21]. Older Bq L^{-1} records are converted with matching rain depth (L m^{-2}); method codes, uncertainty and decay conventions are retained. Collector–grid surface differences enter the representativeness error model. Nondetections are censored observations; missing values are not zeros.

Candidate response vectors will be evaluated using air only, deposition only and both. Known scenarios will generate synthetic measurements with actual gaps, sampling intervals and uncertainty. Recovery tests will include cases generated outside the fitted source assumptions, so recovering a model from itself is not the only check. Measurement and representativeness errors, temporal/spatial dependence and explicit model discrepancy will be included; covariance matters when rain depth is used to convert deposition.

Uncertainty-scaled sensitivity vectors and singular values will screen confounded responses. Low-dimensional likelihood profiles [8] will explore source amplitudes and nuisance terms at each simulated wet setting. Wet-response interpolation is not assumed. The discrimination map will report recovery probability for prescribed contrasts and detectable source-amplitude bounds. A planned reporting criterion is ≥80% recovery at ≤5% false discrimination in synthetic tests, with simulation uncertainty; failure to reach it is reported as a bound outside the tested range, not a fabricated threshold.

Parameters, regime definitions and scoring rules will be fixed before unblinding held-out tracer responses. Separate air/deposition errors, joint predictive scores and interval coverage will be compared with R using whole-year blocks and method-stratified checks. A candidate is supported only if its intervention has a verified budget effect and its improvement transfers across held-out years without a compensating deterioration beyond the pre-specified uncertainty tolerance. Otherwise it is rejected or classified as indistinguishable among the tested alternatives.

Interdisciplinarity, open science and research-content diversity

Environmental radioactivity determines source/decay and collector conventions; transport physics supplies budgets; statistical experiment design tests information content. Their integration is operational: the same source-resolved simulation is passed through the measurement operator and recovery analysis, preventing disconnected modelling and statistical work packages.

A data-management plan by M3 will define provenance, access, licences and retention. The protocol and withheld blocks will be versioned before fitting. Public outputs will include a DOI-archived analysis release, CSV/NetCDF metadata, executable tests and synthetic examples that run without restricted DEHM code or station records. Permitted derived data will accompany manuscripts; private data remain access-controlled. FAIR metadata and open formats support reuse within Europe's open-science ecosystem [20], without implying formal EOSC onboarding.

Sex and gender are not explanatory variables for the project's atmospheric inventories, fluxes or numerical interventions; no human exposure, health outcome or participant data are analysed. Sex-disaggregating these environmental outputs would not address the research question. This assessment will be revisited if the research content changes. Geographic and observational representativeness are addressed through explicit coverage and sampling tests. Inclusive mentoring and collaboration are separate measures in Section 1.3. No new radioactive releases or handling are involved.

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

The proposed host is Aarhus University (AU), Department of Environmental Science, Roskilde, with Zhuyun Ye as main supervisor. Her institutional profile documents regional Eulerian modelling and data assimilation [13]; she is a co-author of the CAMS regional production-system description [18]. The applicant reports three previous one-month research visits and an established joint ^{7}Be–DEHM workflow [16]. This preparation supports rapid progression from implementation to independent experimental design.

Period / direction	Specific competence and verifiable output
M1–M4 AU → Chham	Source-tag coding, boundary accounting and numerical verification under Ye's guidance: reviewed code and a passing reconstruction/closure test suite.
M4–M12 AU ↔ Chham	Controlled process contrasts and focused training in inverse-problem uncertainty: a reviewed synthetic-recovery protocol before real-data fitting; one methods presentation.
M8–M18 Chham → AU	Radionuclide production, decay, collector conventions and sampling-aware R/Python workflows: two practical sessions with reusable exercises and unit tests.
M18–M24 Chham-led integration	Lead the held-out assessment, developer demonstration and open release; produce a documented research portfolio and follow-on funding concept.

Table 5. Training is assessed through executable methods and independent decisions, not attendance alone. A specialist methodological reviewer for uncertainty analysis is a pre-submission hosting item.

Weekly progress meetings and quarterly design reviews will separate technical progress from scientific interpretation. The career-development plan will be agreed in M1 and reviewed every six months. Chham will lead implementation, analysis and documentation; Ye will review DEHM interventions and scientific claims. An independent methodological review of the recovery/scoring protocol is planned before data fitting; its responsible person or training route must be recorded in the hosting plan.

Inclusive practice will include transparent authorship/contribution records, equitable access to presenting and leading tasks, invitations that seek gender-diverse participation in methods sessions, accessible hybrid materials and scheduling compatible with caring responsibilities. Feedback on mentoring and workload will be part of the six-month reviews. These measures concern working conditions and knowledge transfer; they do not substitute for the research-content assessment in Section 1.2.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Essaid Chham is a postdoctoral researcher at Universidad Miguel Hernández. The applicant's record reports a PhD in Chemistry (Granada, 2018) and a PhD in Physics (Abdelmalek Essaâdi, 2019), Erasmus Mundus experience and María Zambrano/APOSTD postdoctoral periods [16]. His combination of Monte Carlo modelling, environmental radioactivity and atmospheric analysis matches a project connecting source physics, numerical tracers and observations.

His research on ^{7}Be prediction [10], sampling-interval effects in CWT/PSCF [11] and Open-AMA software [12] supports measurement-aware evaluation and reproducibility. Existing DEHM, Fortran, R/Python and NetCDF experience, and teaching/supervision reported in his CV [16], provide the implementation base. The fellowship's new competence is defensible process inference and independent model-development leadership, not a repetition of routine data processing.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

By M24, Chham will own a portable diagnostic protocol, lead its scientific interpretation and demonstrate its use to model developers. The career portfolio will contain reviewed code, two targeted lead-author manuscripts, teaching materials and a follow-on concept for a model-development or environmental-assessment programme. Six-month reviews will assess independent design decisions and user feedback. This provides evidence for academic and atmospheric-modelling roles beyond radionuclide analysis; subsequent multi-model work is a career pathway, not an added fellowship obligation.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination will reach atmospheric modellers and environmental-radioactivity researchers; exploitation will prioritise a usable diagnostic method; communication will explain why a model can match measurements for the wrong reasons. Each activity has a defined audience, product and check of usefulness.

Audience / timing	Product and uptake measure
Scientific community M15–M24	Two manuscripts: source-resolved response/budgets and observation-space discrimination/transfer; preprints and open-access routes. At least one conference contribution. Track submissions, DOI releases and scientific feedback, not promised acceptance.
AU/DEHM developers M12 and M22	Demonstrate the benchmark on a documented case and deliver a developer note identifying supported changes, unresolved alternatives and regression tests. Record review comments and reproducibility issues; target one re-run by a user other than the fellow.
Monitoring/data specialists M18–M24	A concise observing-value brief: where paired deposition adds information and when finer sampling would help. Seek DWD and other relevant specialists' feedback; these are intended audiences, not newly committed partners.
Wider research users / public M8–M24	An open worked example, two practical training sessions and an accessible brief on natural radiotracers and model uncertainty. Record attendance, questions and tutorial completion; avoid equating downloads with impact.

Table 6. Deliverables and engagement targets are project commitments; external adoption, attendance and publication acceptance are not guaranteed.

The analysis layer will have explicit input schemas, units, sample-window metadata, tests and a permitted-use licence. A small synthetic benchmark will reproduce the full workflow without confidential inputs, enabling independent review and later adaptation to other models. Versioned releases will separate host-owned model code from reusable analysis tools. User feedback will be logged and addressed before the final release; the development team is the first target user, not a claimed service-wide deployment.

2.3 The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. The project will deliver one verified multi-origin ^{7}Be framework, three matched long-period configurations anchored to the 2009–2024 observational baseline, and a discrimination map indexed by regime, sampling and uncertainty. Additional quality-controlled years will broaden evaluation. It tests whether process explanations transfer across years, rather than only improving a short-window fit. Developers can see whether concentration agreement is purchased at the cost of deposition or altered source assumptions. The reusable observation operator and synthetic tests extend this contribution beyond DEHM.

Technological and environmental-assessment pathway. DEHM is part of the CAMS regional air-quality system [18]. STRAT-TRACE will provide its development community with diagnostic evidence and tests relevant to aerosol transport/removal, then document whether a supported intervention merits further pollutant-specific evaluation. Better-targeted development can reduce uninformative tuning and improve the traceability of model revisions. Developer time, usability and successful independent re-runs will be assessed; forecast improvement must be demonstrated in subsequent operational validation, not assumed from ^{7}Be alone.

Societal and European relevance. More credible aerosol-process evaluation supports the evidence base for cleaner-air and ecosystem assessments, consistent with the EU Zero Pollution Action Plan [19]. The direct fellowship contribution is a transparent diagnostic method and actionable guidance for modellers and monitoring specialists. Potential later benefits depend on validated transfer to the relevant pollutants and assessment systems. FAIR, reusable outputs [20] lower access barriers to this evidence. This is a defined route from methods to environmental services, rather than an unsupported numerical claim about health or economic savings.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

A single critical path links verified labels, controlled source–removal responses and held-out evaluation across years. Computational access is already available to the team. The pre-submission check in Section 3.2 documents this access, input coverage and technical readiness; M4 verifies labels and budgets, not whether computing can first be obtained.

WP / span / effort	Tasks and measurable output	Milestone
WP1 M1–M4 / 3 PM	Implement source/budget tests; inventory existing and incoming records; pre-register temporal blocks. D1: verified source/measurement protocol and data-management plan.	MS1, M4: closure and source-reconstruction gate passed; inference protocol frozen.
WP2 M3–M9 / 5 PM	Run the tagged reference over the common window; verify regime coverage. D2: origin budgets and dataset, anchored to 2009–2024 and extended where inputs support it.	MS2, M9: O1 assessed; source fractions reconstruct totals.
WP3 M7–M15 / 6 PM	Run both wet settings over the same full period; reweight sources and verify intervention budgets. D3: response library and first manuscript draft.	MS3, M15: O2 assessed; no selective omission of held-out cases.
WP4 M12–M21 / 5 PM	Recovery and uncertainty tests; locked temporal/spatial and method-stratified evaluation, including quality-controlled extra years. D4: discrimination map, observing brief and second manuscript draft.	MS4, M21: O3 classified, including explicit non-discrimination bounds.
WP5 M1–M24 / 5 PM	Training, supervision, data stewardship, developer engagement and dissemination. D5: reusable release, methods materials and career portfolio.	MS5, M24: releases complete; two manuscript submissions targeted.

Table 7. Total researcher effort is 24 person-months (PM). Overlapping calendar spans do not imply multiple full-time researchers. WP2 starts with preparation; production runs follow MS1. External consultation does not control core delivery.

WP	1–3	4–6	7–9	10–12	13–15	16–18	19–21	22–24
WP1	●	●						
WP2	●	●	●					
WP3			●	●	●			
WP4				●	●	●	●	
WP5	●	●	●	●	●	●	●	●
Gate / output		MS1	MS2		MS3		MS4	MS5

Gantt chart. Filled cells indicate activity during all or part of the interval. D1–D5 correspond to MS1–MS5; the data-management plan is due in M3. No secondment or placement lies on the critical path.

Execution budget and control of scope

For the 2009–2024 baseline, the programme comprises 48 configuration-years (3 × 16), not a runtime ceiling. A common extension to N years scales this to 3N, plus spin-up and verification. The reduced-chemistry setup and existing computational access are retained. The same archive drives all cases; tags add species, not meteorological integrations. A tagged representative-month test documents CPU time, memory and output. A 20% execution allowance covers restarts and verification for the selected window, without assuming unlimited capacity.

Hourly receptor series and online regional/pressure-band budget sums span the full period. Full-domain high-frequency output is restricted to selected episodes; checkpoints, ledger tests and checksums control storage and reruns. Resolution changes, transport-operator perturbations and extra models remain outside the core. Output reduction must not shorten the common experiment period or remove held-out tests. Effort includes verification, metadata harmonisation, interannual uncertainty analysis and knowledge transfer, not model runtime alone.

Delivery risks and scientific decision branches

Non-identification in a particular regime is a scientific outcome with a detection bound, not an unfinished task. A network that supports no empirical discrimination remains a material limitation. The table therefore separates delivery threats from pre-specified interpretive outcomes; likelihood ratings are qualitative residual assessments after mitigation, not measured probabilities.

Risk / residual likelihood / impact	Trigger, mitigation and retained result
Numerical closure or linearity fails Medium / High	Failed ledger or reconstruction tests stop source-fraction claims. Audit boundary, limiter and decay terms; use a verified coarser partition or source-restricted reruns within the costed reserve. Report the numerical limitation if closure cannot be established.
Paired records or metadata inadequate Medium / High	Audit station gaps and method changes, including pre-/post-2018 strata. Extra deposition years enter only after receipt and quality control; delays do not block baseline delivery. Missing records are not zeros. Inadequate core coverage requires redesign.
Little discrimination across the available network Medium / Medium	Early synthetic tests evaluate the actual design. Report compatible scenarios and contrast bounds, then quantify the value of alternative sampling in the benchmark. Preserve the source/budget results; do not label a unique atmospheric cause.
I/O, archive or workflow delays Medium / High	Use existing computational access, checkpoints, execution allowance and bounded 3-D output. Audit archived inputs and track throughput; prioritise online sums. Preserve all three full-period cases and locked evaluation, rather than impose a four-year runtime cap.
Candidate improvement fails held-out tests Medium / Medium	Separate air/deposition scores and source/discrepancy stress tests expose compensation. Deliver a documented rejected explanation and a regression test, rather than retune on the validation blocks.

Table 8. Qualitative residual assessments will be reviewed at each milestone. The response library and discrimination bounds remain deliverables when a particular explanation is rejected.

3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

AU's Department of Environmental Science develops atmospheric transport, chemistry and deposition models across scales [14,15]. Its DEHM contribution to CAMS [18], together with the joint ^{7}Be implementation [7], gives the proposed host a direct scientific and user-community fit. The research configuration is not assumed to be identical to operational CAMS settings; transfer will concern tested diagnostics before any operational model change.

Existing computational access and the team's reduced-chemistry DEHM workflow support day-1 execution without a new supercomputing award. Hosting combines the code/input workflow, secure storage, workspace and onboarding with weekly supervision and quarterly code/design review. Chham leads delivery; Ye guides DEHM science and coordination. Licensing, station-data permissions, storage provision and a methodological review route will be documented before submission. No new field campaign is required.

PRE-SUBMISSION DOCUMENTATION — remaining items for this review draft. Record computational access, infrastructure/contact and tagged-run performance; specify storage, code/data permissions, supervision and uncertainty-training support. Complete the station-period/method inventory, separating the available baseline from years being obtained, and the wet-operator test. The applicant has confirmed computational access and the lightweight configuration; this does not replace institutional documentation.

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